

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and implemented control logic for **automation equipment** and machinery integration projects, ensuring seamless operation and adherence to project specifications.
- Configured and tested **HMI interfaces** and basic robotic system parameters to support the deployment of automated manufacturing solutions.
- Integrated **machine vision solutions** using Cognex platforms for automated quality control, achieving defect detection rates of approximately **98.5%** on sample production runs.
- Contributed to the development of **SCADA and MES integration** components, focusing on data collection and visualization for enhanced operational oversight.
- Troubleshoot and optimized **industrial control systems**, resolving an average of **3-4 critical issues** per month to maintain system reliability and performance.
- Developed and delivered technical demonstrations and training materials for **digital manufacturing technologies** to DMI colleagues and external clients, enhancing team knowledge transfer.
- Containerized control system applications using **Docker** and managed deployments on **AWS EC2 instances**, improving deployment consistency by **30%**.

SKILLS

Industrial Control Systems – PLC Programming, HMI Design, SCADA Integration, MES Concepts, IEC 61131

Machine Vision – Cognex, Keyence, Halcon, Quality Control, Defect Detection

Automation & Robotics – Equipment Integration, Automation Solutions, Robotic Systems

Cloud & DevOps – AWS EC2, Docker, CI/CD, Jenkins, Git

Protocols & Connectivity – Ethernet/IP, Profinet, EtherCAT, Modbus, OPC-UA, MQTT

Programming Languages – Python, Java, SQL

Data & Analytics – Power BI, Tableau, DAX, Power Query, Data Modelling

PROJECTS

Automated Quality Inspection System (HCL Technologies)

- Developed and deployed a **machine vision system** for an enterprise client's production line, integrating **Cognex** hardware to automate visual inspection tasks.
- Implemented Python scripts to process image data and identify defects, reducing manual inspection time by approximately **40%** and improving accuracy.
- Configured system parameters and created **HMI interfaces** for operators to monitor inspection results and adjust sensitivity settings as needed.

Industrial Data Acquisition & Visualization (HCL Technologies)

- Engineered a data pipeline using **Python** and **SQL** to collect real-time operational data from manufacturing equipment via **OPC-UA**.
- Built interactive **Power BI dashboards** to visualize key performance indicators, such as machine uptime, cycle times, and error rates, for management review.
- Managed data flow and storage within a **PostgreSQL** database, ensuring data integrity for historical analysis and reporting.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)